

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 March 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Image Upload	Image uploading by users can be done via web interface.
		Image formats that will be accepted by the system include PNG, JPG, JPEG.
FR-2	Image Preprocessing	System converts uploaded image in RGB format.
		Image is resized depending on the size of the input model.
		Image pixel values are normalized into system values.
FR-3	Malaria Detection	System examines uploaded image with established deep learning model.
		System identifies image as Parasitized or Uninfected.
FR-4	Prediction Result Display	System displays to the user the prediction result.
		System has confidence score of prediction.
FR-5	Model Selection	System loads the trained model (malariadetectorfinal.h5) which is the best.
		Prediction on the loaded model is done by system.
FR-6	API Prediction Service	System suggests POST /predict API to predict images.
		The API gives an output of the prediction in the form of a YN.
FR-7	System Health Monitoring	GET /health API is provided in the system.
		System is used to indicate that the model is loaded and system is running.
FR-8	Batch Prediction	System is able to predict several images in a folder.
		Prediction summary on batch images is generated by system.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional requirement	Description
NFR-1	Usability	The system must allow easy and friendly web interface to post images and view predictions.
NFR-2	Security	The system must be able to validate files uploaded to minimize abuse of the files and only allow files compatible with the system.
NFR-3	Reliability	The trained deep learning model should be able to generate correct predictions every time the system is used.
NFR-4	Performance	The system must be able to produce the prediction results in a couple of seconds when the image is uploaded.
NFR-5	Availability	The web application must be in place and functional as long as the server is up and running.
NFR-6	Scalability	The system must support a variety of prediction requests and should be capable of supporting the growth of databases or model enhancements in the future.